

EventShift: Reliability-Aware Residual Adaptation for Day–Night RGB–Event Semantic Segmentation

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Abstract. Event cameras preserve motion and edge cues under low light, but their sparse change-driven responses are not uniformly reliable when RGB appearance is already strong. We study this day–night trust trade-off in the EBMV@ECCV 2026 Event-Guided Semantic Segmentation with CoSEC challenge track and introduce EventShift, an RGB-anchored residual adapter combining an image-level low-light budget, reliability-aware spatial allocation, RMS-bounded updates, and a multi-window edge prior. Relative to a matched RGB baseline on our disclosed CoSEC split, EventShift improves night mIoU by 4.58 points with a 0.16-point day decrease, raising the day–night average by 2.21 points. A separate composite challenge system based on the same core residual-adaptation principle placed second on the official challenge Test Phase leaderboard with 41.11 mIoU. In a separate internal DSEC11 experiment, staged event adaptation improves mIoU from 78.36 to 79.70. These results support the use of bounded, reliability-aware event correction under the evaluated protocols.

Keywords: Event-based vision · Semantic segmentation · Multimodal perception · Illumination variation

1 Introduction

Event cameras asynchronously record polarity changes with high temporal resolution, high dynamic range, and little motion blur [8]. Their sparse streams preserve moving boundaries when frames are dark or blurred, but omit absolute appearance and also respond to ego-motion, flicker, and sensor noise [8]. They therefore complement RGB rather than replace it.

This complementarity makes day–night segmentation an asymmetric fusion problem. In Fig. 1, transferred MambaSeg and BRENet checkpoints achieve substantially higher night than day mIoU on CoSEC. When nighttime RGB appearance is weak, moving boundaries from events become relatively useful; by day, RGB carries fine-grained appearance, while event features introduced before stable semantics form can inject ego-motion, flicker, and noise. Because

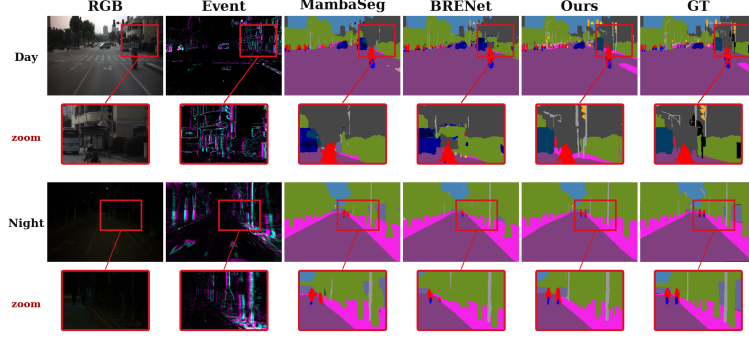


Fig. 1: CoSEC examples from transferred MambaSeg and BRENet checkpoints and our full EventShift checkpoint. The transfers reveal contrasting day–night operating points; controlled ablations are reported in Table 1.

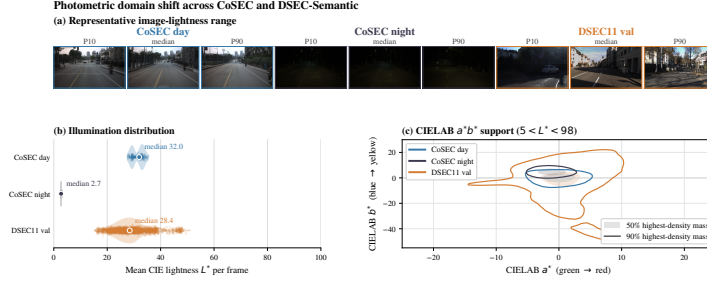


Fig. 2: Photometric shift across CoSEC and DSEC-Semantic after sRGB-to-CIELAB (D65) conversion. (a) Representative frames by lightness percentile. (b) Per-frame mean L^* distribution. (c) 50%/90% highest-density a^*b^* regions from 2-D histograms.

the transfers differ in architecture and training, this pattern is design motivation rather than causal evidence.

Figure 2 quantifies the condition shift: median per-frame mean CIELAB lightness is 32.0 for CoSEC day, 2.7 for CoSEC night, and 28.4 for DSEC11 validation. This separation motivates condition-dependent fusion. EventShift keeps RGB as the semantic anchor and inserts RMS-bounded residuals at `res3/res4`, with image appearance controlling frame-level strength and reliable multi-window event structure controlling spatial allocation.

Our contributions are:

- We introduce RGB-anchored residual adaptation whose image-level low-light budget controls frame-level strength and whose sample-wise RMS ceiling caps the magnitude of event-induced changes to pretrained RGB features.
- We derive a deterministic reliability/support proxy and reliability-weighted MW edge prior that steer spatial allocation toward coherent, supported event structure.
- On our disclosed CoSEC split, EventShift improves night mIoU by 4.58 points with a 0.16-point day decrease. A separate composite system based

on the same principle placed second on the official challenge Test Phase leaderboard with 41.11 mIoU.

2 Related Work

Driving segmentation and event benchmarks. Cityscapes established the standard urban-scene taxonomy, while Mask2Former provides the masked-attention segmentation backbone used here [3, 4]. ACDC and DANNet expose the degradation of RGB segmentation under adverse conditions and day–night shift [19, 21]. DSEC-Semantic introduced event-camera driving labels [6, 9, 20]; CoSEC adds synchronized coaxial RGB–event data [18] and supports the EBMV@ECCV 2026 semantic-segmentation challenge studied here [7].

Event and multimodal semantic segmentation. EV-SegNet and ESS established event-only segmentation and image-to-event transfer of supervision [1, 20]. MambaSeg uses dual state-space encoders, BRENet performs flow-guided registration, and ESC combines edge-aware re-coding with uncertainty optimization [2, 10, 22]; related RGB-X work emphasizes cross-modal rectification [23]. Drawing on residual learning, adapters, and channel gating [11–14], EventShift instead bounds and spatially allocates event-induced adaptation to a pretrained RGB path.

3 Method

3.1 Overview

Given an RGB frame I_t and surrounding events, the RGB backbone produces multi-scale semantic anchors F_ℓ . The event representation and its reliability/MW priors guide bounded residual adapters at **res3/res4**:

$$F'_\ell = F_\ell + U_\ell, \quad (1)$$

where U_ℓ is the event-induced update. The corrected pyramid enters the standard Mask2Former pixel and transformer decoders; Sec. 3.6 defines the semantic output and supervision. Figure 3 summarizes the complete path.

3.2 Bidirectional Event Representation

For each RGB frame, the manifest provides an event H5 file and two temporal windows: an old window before the frame and a new window after the frame. For a window $w \in \{-, +\}$ with events (x_i, y_i, t_i, p_i) , we set $\sigma_i = 1$ for positive polarity and $\sigma_i = -1$ otherwise, then map timestamps to a normalized bin coordinate

$$u_i = (B - 1) \frac{t_i - t_0^w}{\max(t_1^w - t_0^w, 1)}, \quad l_i = \lfloor u_i \rfloor, \quad r_i = \min(l_i + 1, B - 1). \quad (2)$$

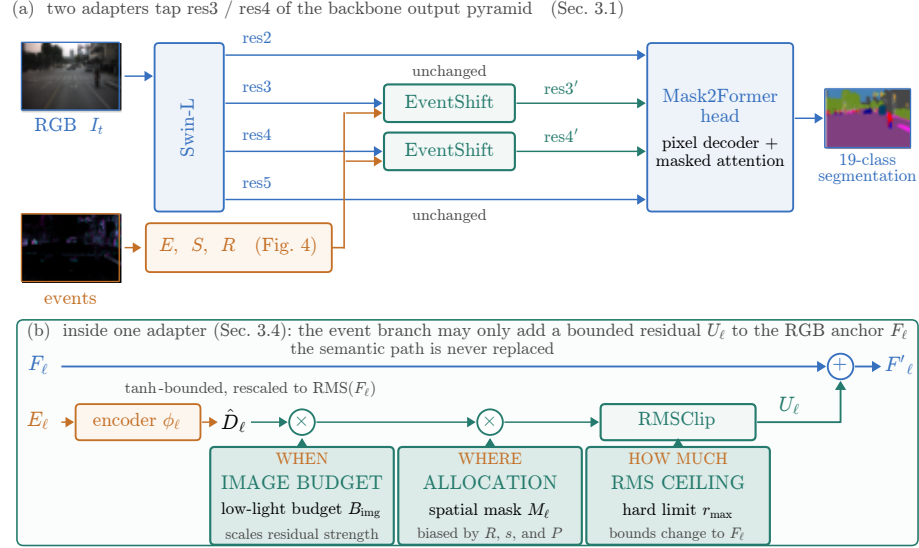


Fig. 3: Architecture of EventShift. Adapters correct **res3/res4**, while **res2/res5** remain unchanged. Inside each adapter, RGB remains the semantic anchor and events add only U_ℓ : B_{img} controls frame-level correction strength, M_ℓ controls *where* it is allocated, and r_{max} controls *how much* it may change F_ℓ .

Each event is linearly interpolated between adjacent temporal bins in a polarity-aware voxel grid with $B = 5$ bins,

$$V_k^w(y_i, x_i) += \sigma_i((1 - \alpha_i)\mathbf{1}[k = l_i] + \alpha_i\mathbf{1}[k = r_i]), \quad \alpha_i = u_i - l_i. \quad (3)$$

At the same pixels we accumulate the event count $n^w(y_i, x_i) += 1$ and polarity counts p^+ and p^- . The old and new voxels are concatenated into a 10-channel bidirectional tensor,

$$E = \text{concat}(V_1^-, \dots, V_B^-, V_1^+, \dots, V_B^+). \quad (4)$$

We standardize all nonzero entries using one sample-wise mean and standard deviation over the concatenated tensor, leaving zeros unchanged. This retains pre- and post-frame context. The voxel and main reliability paths use 50 ms before and after the RGB timestamp, while the MW edge prior uses symmetric 25/50/100 ms half-windows (Sec. 3.5); full EventShift therefore requires at most 100 ms of look-ahead. The public challenge scores submitted masks using segmentation metrics, with mIoU as the primary metric, and specifies neither a latency metric nor a causal constraint [7]; we therefore evaluate an offline protocol with fixed temporal look-ahead. The look-ahead precludes strictly causal deployment, for which the new window and future MW support must be replaced by past-only events. Figure 4 summarizes the event-processing path.

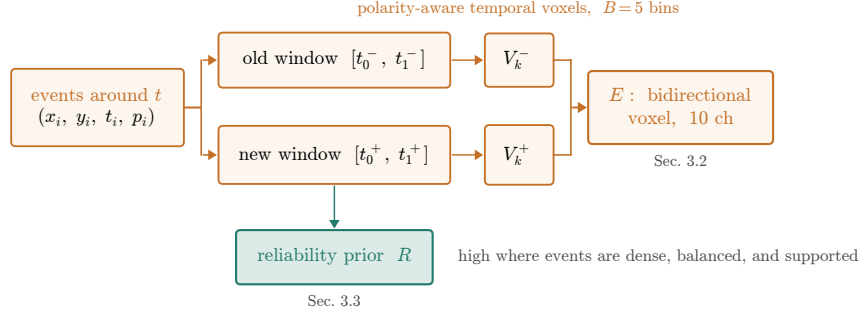


Fig. 4: Event representation and reliability-prior construction. The events around t are split into an old and a new window, each converted into a polarity-aware temporal voxel; concatenating the two gives the 10-channel bidirectional tensor E . The same windows also yield the scalar reliability prior R , which is high only where events are dense, balanced in time and polarity, and supported. Definitions are given in Sec. 3.2 and Sec. 3.3.

3.3 Reliability Prior

Events should be trusted only where they provide sufficient and balanced support. Alongside the voxel tensor, we compute the old-window event count n^- , the new-window event count n^+ , the positive event count p^+ , and the negative event count p^- . These statistics define density, temporal balance, and polarity balance:

$$d = \log(1 + n^- + n^+), \quad (5)$$

$$b_t = 1 - \frac{|n^- - n^+|}{n^- + n^+ + \epsilon}, \quad (6)$$

$$b_p = 1 - \frac{|p^+ - p^-|}{p^+ + p^- + \epsilon}, \quad (7)$$

where ϵ is a small positive constant. We also compute a support mask from a density percentile and temporal-balance threshold,

$$s = \text{Dilate}_{r_s} (\mathbf{1} [d > \tau_\rho(d \mid d > 0)] \cdot \mathbf{1} [b_t > \tau_t]), \quad (8)$$

where τ_ρ is the configured nonzero-density percentile and r_s is the support-dilation radius. The reliability prior is

$$R = s \cdot \bar{d}^{\gamma_d} \cdot b_t^{\gamma_t} \cdot b_p^{\gamma_p}, \quad (9)$$

where $\bar{d}$ is stage-wise max-normalized density. We use $S = [\bar{d}, b_t, b_p, s]$ as the four-channel *event-statistics tensor* and R as a deterministic scalar reliability proxy, not a calibrated probability of event correctness. Both are resized to each feature stage; S_ℓ conditions A_ℓ , while R_ℓ and s_ℓ provide reliability and support biases. Exponents are given in Sec. 4.3.

This prior is intentionally more selective than raw event density: daytime CoSEC frames can contain many events on building edges, lane markings, vehicles, and thin poles, but only regions with sufficient density and temporal consistency receive support. EventShift therefore treats events as reliability-gated residual evidence rather than as a dense semantic image.

3.4 Reliability-Aware Residual Adaptation

The adapter first maps the event tensor to the feature resolution and predicts a residual direction, then normalizes it by the RGB feature scale so the event branch cannot inject arbitrary feature magnitudes:

$$D_\ell = \tanh(\phi_\ell(E_\ell)); \quad \hat{D}_\ell = \frac{\text{RMS}(F_\ell)}{\text{RMS}(D_\ell) + \epsilon} D_\ell. \quad (10)$$

At each injection stage, ϕ_ℓ uses a 3×3 10-to-128 convolution, GN(32), GELU, another 3×3 128-to-128 convolution, GN(32), GELU, and a 1×1 128-to- C_ℓ projection; $C_\ell=384/768$ at **res3/res4**. A_ℓ is a 1×1 convolution from $2C_\ell+4$ channels to one logit, followed by the biases and softplus normalization below. The final ϕ_ℓ projection and A_ℓ weights are zero-initialized (allocator bias 0); preceding layers use PyTorch defaults. A learned allocator predicts the spatial distribution of residual updates. Reliability and support enter as explicit biases; the optional MW structural prior P_ℓ , constructed in Sec. 3.5, biases only this allocator:

$$q_\ell = A_\ell([F_\ell, \hat{D}_\ell, S_\ell]) + \beta_r(2R_\ell - 1) + \beta_s(2s_\ell - 1) + \beta_P(P_\ell - 1), \quad (11)$$

$$M_\ell = \text{clip} \left(\frac{\text{softplus}(q_\ell)}{\text{mean}(\text{softplus}(q_\ell)) + \epsilon}, 0, m_{\max} \right), \quad (12)$$

Here S_ℓ denotes the resized event-statistics tensor. The reliability/support switch in Table 1 enables the β_r and β_s terms, while the MW column controls β_P . Because the image budget and RMS ceiling are bundled throughout the residual family, Table 1 should not be interpreted as independently validating each term in this decomposition.

Before spending any event residual, we apply an image-level low-light budget as a fixed guardrail. From RGB luma Y and local contrast $c = |Y - \text{avg}_{7 \times 7}(Y)|$, we compute

$$L = \text{mean} \left[\text{clip} \left(0.7 \frac{\tau_Y - Y}{\tau_Y} + 0.3 \frac{\tau_c - c}{\tau_c}, 0, 1 \right) \right], \quad (13)$$

$$B_{\text{img}} = B_{\min} + (1 - B_{\min}) \text{sigmoid}(b_L + g_L L). \quad (14)$$

This budget reduces event residual strength in bright, high-contrast frames and keeps more capacity for low-light frames, where RGB evidence is less reliable. It is held fixed in all allocation ablations. The final update is

$$U_\ell = \text{RMSClip}(\lambda_{\max} B_{\text{img}} M_\ell \hat{D}_\ell, F_\ell, r_{\max}). \quad (15)$$

For a sample $X_j \in \mathbb{R}^{C \times H \times W}$, RMS is computed over channel and spatial dimensions, not across the batch:

$$\text{RMS}(X_j) = \sqrt{\frac{1}{CHW} \sum_{c,h,w} X_{j,c,h,w}^2}, \quad (16)$$

$$\text{RMSClip}(X_j, F_j, r) = X_j \min\left(1, \frac{r \text{RMS}(F_j)}{\text{RMS}(X_j) + \epsilon}\right). \quad (17)$$

Unlike LayerNorm or BatchNorm, which normalize activations without imposing a residual-to-anchor bound, RMSClip preserves the residual direction and enforces $\text{RMS}(U_j) \leq r_{\max} \text{RMS}(F_j)$ independently for each sample. This hard energy bound is independent of batch statistics and prevents sparse or noisy event responses from overwhelming the RGB anchor; concrete settings are given in Sec. 4.3.

3.5 Multi-Window Prior for Residual Allocation

For the MW prior, $a \in \{25, 50, 100\}$ ms denotes the half-window radius around the RGB timestamp. We count n_a^- events in $[t - a, t]$ and n_a^+ in $[t, t + a]$, corresponding to full symmetric durations of 50, 100, and 200 ms. Short windows preserve recent boundaries, while longer windows provide denser support and set the maximum look-ahead to 100 ms. At radius a , density and temporal/polarity balance are

$$\begin{aligned} d_a &= \log(1 + n_a^- + n_a^+), \\ b_{t,a} &= 1 - \frac{|n_a^- - n_a^+|}{n_a^- + n_a^+ + \epsilon}, \\ b_{p,a} &= 1 - \frac{|p_a^+ - p_a^-|}{p_a^+ + p_a^- + \epsilon}. \end{aligned} \quad (18)$$

With plain edge map $e_a := \bar{d}_a$, support s_a , and $R_a = s_a \bar{d}_a^{1/2} b_{t,a} b_{p,a}^{0.25}$, the reliability-weighted prior is

$$\tilde{P} = \frac{\sum_a e_a R_a}{\sum_a R_a + \epsilon}, \quad P = \text{clip}\left(\frac{\tilde{P}}{\text{mean}(\tilde{P}) + \epsilon}, 0, 3\right). \quad (19)$$

For the +MW edge-prior ablation, the density–polarity edge score at radius a is $q_a = d_a(0.5 + 0.5b_{p,a})$, with $\bar{q}_a = q_a / \max_{x,y} q_a$. A bar denotes division by the spatial maximum, with an all-zero map left at zero. The ablation uses the unweighted channel mean of the per-radius normalized density $\bar{d}_a$, normalized score $\bar{q}_a$, and polarity-balance map $b_{p,a}$, followed by the same spatial mean normalization and clipping.

The two single-prior paths are not algebraically identical, but they share a monotone 50 ms core. For positive $s_{50}, b_{t,50}, \bar{d}_{50}, b_{p,50}$, holding support and

temporal balance fixed gives

$$\begin{aligned} \frac{\partial R_{50}}{\partial \bar{d}_{50}} &= \gamma_d \frac{R_{50}}{\bar{d}_{50}} \geq 0, & \frac{\partial R_{50}}{\partial b_{p,50}} &= \gamma_p \frac{R_{50}}{b_{p,50}} \geq 0, \\ \frac{\partial q_{50}}{\partial d_{50}} &= \frac{1}{2}(1 + b_{p,50}) > 0, & \frac{\partial q_{50}}{\partial b_{p,50}} &= \frac{1}{2}d_{50} \geq 0. \end{aligned} \quad (20)$$

Spatial normalization preserves this ordering before clipping saturates. Both paths therefore favor dense, polarity-balanced locations in the shared 50 ms window; reliability/support additionally rejects temporally inconsistent or unsupported activity, while MW aggregates three radii. Equation (20) establishes aligned dependence on shared cues, not equivalence of the priors. Full EventShift uses the reliability-weighted prior in Eq. (19); it biases the residual allocator but never replaces the RGB feature.

Figure 5 connects these controls to the corrected feature: in the darker example, a larger image budget coincides with concentrated reliability and a stRong-Liner residual.

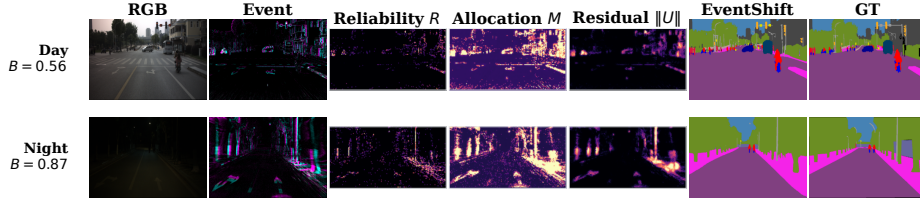


Fig. 5: Measured EventShift feature behavior on representative CoSEC day and night cases (heatmaps share a scale per cue). The low-light budget rises from $B_{\text{img}} = 0.56$ (day) to 0.87 (night), and the **res3** residual-to-RGB RMS ratio rises from 4.1% to 7.1%. Reliability suppresses unsupported activity while the allocator concentrates the residual on supported structure.

Having specified where and how strongly events may modify the pyramid, we next describe how the corrected features produce the final semantic map and training signal.

3.6 Semantic Output and Training Objective

Replacing **res3/res4** by their corrected features yields a four-level pyramid $\mathcal{F}'$. The Mask2Former decoder returns $Q = 100$ class logits $c_q \in \mathbb{R}^{K+1}$, including no-object, and mask logits $m_q(x)$. For semantic class k , inference combines query classification and masks as

$$z_k(x) = \sum_{q=1}^Q \text{softmax}(c_q)_k \sigma(m_q(x)), \quad \hat{Y}(x) = \arg \max_{k \in \{1, \dots, K\}} z_k(x). \quad (21)$$

We resize z to the input resolution before the pixelwise argmax. Thus F'_ℓ is an intermediate representation and $\hat{Y}$ is the submitted semantic map.

Training follows Mask2Former set prediction [3]. We retain its standard matching and loss weights without task-specific tuning: Hungarian matching minimizes $2\mathcal{L}_{\text{cls}} + 5\mathcal{L}_{\text{mask}} + 5\mathcal{L}_{\text{dice}}$ using 12,544 sampled mask points. The supervised mask losses use 12,544 points selected with 3.0 oversampling and a 0.75 importance-sampling ratio. Let $p_q = \text{softmax}(c_q)$, $\tilde{m}_q = \sigma(m_q)$, and y_q be the matched class or $\emptyset$ for an unmatched query. Classification uses

$$\mathcal{L}_{\text{cls}} = -\frac{\sum_{q=1}^Q w_{y_q} \log p_q(y_q)}{\sum_{q=1}^Q w_{y_q}}, \quad w_{\emptyset} = 0.1, \quad (22)$$

so unmatched queries are supervised as no-object. For each matched query and binary target mask g , the point-sampled mask terms are

$$\mathcal{L}_{\text{mask}} = -\frac{1}{|\mathcal{P}|} \sum_{x \in \mathcal{P}} [g(x) \log \tilde{m}_q(x) + (1 - g(x)) \log(1 - \tilde{m}_q(x))], \quad (23)$$

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2 \sum_{x \in \mathcal{P}} \tilde{m}_q(x) g(x) + 1}{\sum_{x \in \mathcal{P}} \tilde{m}_q(x) + \sum_{x \in \mathcal{P}} g(x) + 1}. \quad (24)$$

These terms are averaged over matched masks, and the supervised prediction-stage objective is

$$\mathcal{L}_{\text{seg}} = 2\mathcal{L}_{\text{cls}} + 5\mathcal{L}_{\text{mask}} + 5\mathcal{L}_{\text{dice}}. \quad (25)$$

Deep supervision applies (25) to the final and all auxiliary decoder predictions. We use no event-specific auxiliary loss; segmentation gradients train the decoder, event encoder, and residual adapters end to end.

4 Experiments

4.1 Datasets and Taxonomies

CoSEC provides synchronized coaxial RGB-event recordings for our primary benchmark [18]. The EBMV@ECCV 2026 release contains 2,569 labeled samples across 14 sequences and 982 hidden-label test samples, but no labeled validation partition [7]. We use its official Cityscapes-compatible 19-class format [4]. Our controlled validation split holds out all 257 frames of `Day_Campus_000` and frames 000069–000138 (70) of `Night_Campus_002`; frames 000066–000068 and 000139–000141 are excluded as three-frame boundary gaps. The nearest training and validation frame centers are 401.674 and 402.808 ms apart; after accounting for the widest ± 100 ms MW windows, event support remains separated by 201.674 and 202.808 ms. The resulting split contains 327 validation and 2,236 training samples, each with RGB, a semantic label, event H5 data, and old/new 50 ms windows around the RGB timestamp.

DSEC is a secondary evaluation with a separate 11-class taxonomy that merges fine-grained Cityscapes categories into background [6, 9, 20]; we do not mix it with CoSEC19.

4.2 Protocol and Baselines

We report overall/day/night mIoU on the challenge’s 19-class format [4, 7]. Because validation contains 257 day but only 70 night samples, pooled aggregation is day-dominated; for condition-balanced analysis and checkpoint selection, we therefore also report

$$\text{D-N Avg.} = \frac{\text{Day mIoU} + \text{Night mIoU}}{2}. \quad (26)$$

Classes absent from both ground truth and predictions are omitted when computing mean IoU; a ground-truth-absent class with predicted pixels contributes zero IoU. Overall mIoU remains the official pooled metric. Lacking public CoSEC19 checkpoints, we initialize MambaSeg and BRENet from released DSEC11 weights, replace their heads, and fine-tune on the same split/evaluator. These transfers provide context; Table 1 remains the controlled evidence.

4.3 Implementation Details

All EventShift and RGB-control runs use Swin-L [16] Mask2Former [3] with AdamW [17] (poly decay, weight decay 0.05), gradient clipping at 0.01, mixed precision, and seed 2026. Training applies ImageNet [5] normalization, SSD color augmentation, 512×1024 crops, and short sides $\{512, 624, 768\}$; validation uses short side 624 without test-time augmentation. Event inputs use bidirectional 50 ms windows with five bins, nonzero standardization, event/voxel-bin dropout 0.20/0.10, and 25/50/100 ms MW half-windows.

The shared checkpoint is Cityscapes19-pretrained Swin-L Mask2Former fine-tuned on CoSEC for 4 epochs at 5×10^{-6} and 15 at 10^{-6} ; the checkpoint with the highest overall mIoU defines RGB continuation and initializes every event row in Table 1. Only the event encoders and residual allocators at **res3/res4** are randomly initialized; the RGB backbone and complete segmentation head are retained. We warm the new modules together with the retained head for 2 epochs while freezing the backbone, allowing the decoder to adapt to the introduced event path, then fine-tune all parameters for 13 epochs (batch 3, learning rate 10^{-6} , backbone multiplier 0.01). At both sites, $(\lambda_{\max}, m_{\max}, r_{\max}) = (0.03, 3.0, 0.08)$. We use $B_{\min}=0.20$, $(\tau_Y, \tau_c) = (0.35, 0.08)$, $(b_L, g_L) = (-2, 4)$, $\beta_r=\beta_s=\beta_P=0.5$, and $(\gamma_d, \gamma_t, \gamma_p) = (0.5, 1, 0.25)$. Support uses the 50th nonzero-density percentile, temporal threshold 0.05, and dilation radius 1. The adapter adds 0.47M parameters (0.22% of 215.93M).

4.4 Matched-Backbone Residual Ablation

Table 1 is the primary matched comparison: all rows share the Swin-L Mask2Former backbone, RGB checkpoint, split, and evaluator; each event row applies its variant-specific 2-epoch warmup with the RGB backbone frozen before full fine-tuning. Metrics come from each row’s highest D–N Avg. checkpoint.

Input concatenation tests unfiltered event exposure by appending the 10-channel voxel tensor to RGB; its expanded patch embedding preserves RGB weights, zero-initializes event channels, and warms for 2 epochs. The naive adapter shares the event encoder and **res3/res4** sites but uses an alpha-gated update without budget, RMS control, allocator, or priors. Lower-block rows share the encoder, sites, image budget, and RMS ceiling while toggling reliability/support and MW; because budget and RMS remain bundled, the table is not factorial.

Table 1: Matched CoSEC ablation on one split; the lower panel compares MW-only and rel./support-only per-class gain vectors against the bundled residual.

Variant	Event use	Rel./sup.	MW edge	Overall	Day	Night	D–N Avg.
<i>Fusion controls</i>							
RGB continuation	none	–	–	55.15	55.03	46.81	50.92
Input RGB+event concat.	input	off	off	52.20	52.13	46.61	49.37
Naive event adapter	direct	off	off	52.11	51.97	47.13	49.55
<i>Bounded residual family</i>							
Bundled residual	bounded	off	off	55.15	54.93	47.77	51.35
+ MW edge prior	bounded	off	unweighted	55.03	54.84	50.98	52.91
+ rel./support bias	bounded	allocation	off	55.09	54.86	51.36	53.11
Full EventShift	bounded	allocation	rel.-weighted	55.10	54.87	51.39	53.13
<i>MW-only vs. rel./support-only per-class agreement</i>							
Subset	Gain corr. r		Same direction		Mean $ \Delta\text{IoU} $		
Day	0.978		16/16		0.069		
Night	0.547		7/10		0.515		

Input concatenation and the naive adapter trail RGB continuation, whereas the bundled residual returns to an operating point close to RGB continuation. Within the bounded-residual family, MW-only reaches 50.98 night mIoU and 52.91 D–N Avg.; reliability/support-only reaches 51.36 and 53.11, respectively; full EventShift reaches 51.39 and 53.13. Relative to RGB continuation, full EventShift changes overall and day mIoU by -0.05 and -0.16 points, respectively, improves night mIoU by 4.58 points, and raises D–N Avg. by 2.21 points. Against the bundled residual, MW and reliability/support separately add 1.56 and 1.76 D–N Avg. points, while their combined gain is 1.78 points.

The per-class panel separates overlap from similar aggregate scores. By day, the two single-prior gain vectors have $r=0.978$, the same sign for all 16 present classes, and only 0.069 mean absolute IoU difference. At night, $r=0.547$ and 7/10 signs agree, showing only partial overlap. The bundled residual activates

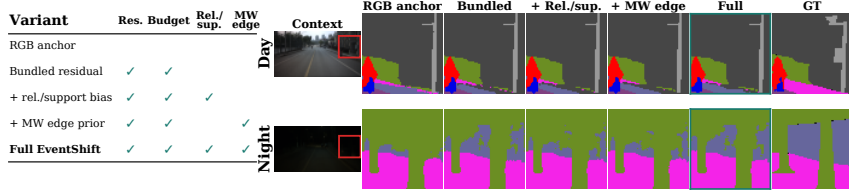


Fig. 6: Matched visual analysis of the bounded-residual rows in Table 1. Budget denotes the image-level low-light budget; Rel./sup. denotes the reliability/support bias. MW uses 25/50/100 ms half-windows.

14 classes in the night evaluator, whereas each single-prior checkpoint activates 13: each suppresses false-positive coverage of one ground-truth-absent class. This shared type of false-positive suppression explains much of the aggregate night gain, while the remaining class-wise differences show that the priors are not interchangeable.

Figure 6 makes the same switches locally inspectable on one day and one night region.

4.5 Cross-Architecture Context

Having established the matched comparison, Table 2 broadens the context to our CoSEC transfers of MambaSeg and BRENet under the same split and evaluator. These architectures provide complementary operating points, while Table 1 remains the controlled evidence.

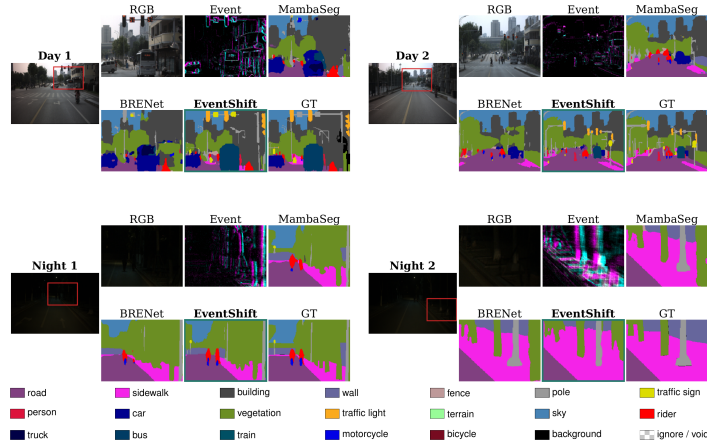


Fig. 7: Illustrative CoSEC day/night comparisons of transferred MambaSeg [10], BRENet [22], full EventShift, and ground truth. Marked regions illustrate the class and boundary differences analyzed in Sec. 4.7.

Table 2: CoSEC19 transfer comparison under a shared split and evaluator. MambaSeg and BRENet start from DSEC11 checkpoints with new 19-class heads; full EventShift starts from the matched Swin-L RGB checkpoint.

Method	Backbone	Overall	Day	Night	D–N Avg.
MambaSeg [10]	VMamba-T	36.30	34.16	56.14	45.15
BRENet [22]	MiT-B2	35.84	32.50	51.20	41.85
Full EventShift	Swin-L	55.10	54.87	51.39	53.13

Under this transfer protocol, full EventShift yields the highest overall and D–N average among the three rows. MambaSeg and BRENet obtain higher night scores than day scores, showing that distinct RGB-event pipelines can occupy markedly different day–night operating points under the shared evaluator.

4.6 Final Challenge Test Result

Beyond controlled validation, a separate composite challenge system based on the same core principle placed second on the official Test Phase leaderboard with 41.11 mIoU [7]. The Test Phase also includes a REAL (Raw and Event Acquired in Low-light) subset [7, 15], so its aggregate score is not directly comparable to the CoSEC-only results in Tables 1 and 2. Because the composite implementation differs from the single-model EventShift formulation, we report the ranking as challenge context rather than controlled evidence.

Table 3: Official challenge Test Phase leaderboard, ranked by mIoU. mAcc and aAcc denote mean class accuracy and overall pixel accuracy, respectively; scores are percentages, and our entry is bold.

Rank	Entry	mIoU	mAcc	aAcc
1	Other	42.21	54.90	90.99
2	Ours	41.11	55.28	89.73
3	Other	37.43	49.65	89.86
4	Other	31.90	41.05	87.97
5	Other	26.83	36.78	86.85

4.7 Class-Level Failure Analysis

The transferred baselines recover the major layout classes but miss several small, appearance-heavy daytime categories. For traffic light, person, and bus, full EventShift reaches 67.0/53.4/85.5 IoU, versus 0.0/10.3/11.5 for MambaSeg and 0.0/8.6/8.6 for BRENet. Traffic lights account for 0.240% of day ground-truth pixels, yet neither baseline predicts them; buses account for 0.512% of ground truth but only 0.074% and 0.058% of the MambaSeg and BRENet predictions, respectively. These deficits are class-concentrated rather than uniform. Event support averages 33.5% by day and 47.3% at night.

Figure 7 shows two day and two night cases; marked regions are illustrative local differences rather than representative samples.

4.8 Additional Evaluation on DSEC-Semantic 11-Class

To test staged adaptation under another taxonomy, we separately train DSEC11 (not zero-shot) on 00_a, 01_a, 02_a, 04_a, and 05_a (4,309 frames), and validate on 06_a, 07_a, and 08_a (3,773 frames). Figure 2 shows its photometric relation to CoSEC; Table 4 reports internal validation only.

DSEC11’s median per-frame mean CIELAB lightness (28.4) is closer to CoSEC day (32.0) than night (2.7), so this experiment is not intended to reproduce the severe night shift. Instead, it tests whether staged residual adaptation can improve a strong RGB solution under another taxonomy and protocol.

Table 4: Internal DSEC11 validation from one RGB initialization; EventShift uses staged rel.-weighted MW adaptation.

Method	Setting	mIoU	pAcc.
RGB baseline	best RGB-only checkpoint	78.36	95.33
EventShift adaptation	staged rel.-weighted MW	79.70	95.73

An RGB-only continuation under the same schedule does not exceed the 78.36-mIoU checkpoint, whereas staged event adaptation reaches 79.70 mIoU and 95.73 pAcc., gains of 1.34 and 0.40 points. This gain is not directly comparable to the CoSEC night result because the datasets, taxonomies, and photometric regimes differ. Because DSEC11 is trained separately on an internal split, the result provides evidence for the portability of staged adaptation rather than zero-shot generalization.

DSEC11 uses Cityscapes19-pretrained Swin-L Mask2Former with compatible weights mapped to the 11-class head. Both rows share a 5,000-iteration RGB stage (about 20 epochs; AdamW, batch 1, learning rate 10^{-5} , backbone multiplier 0.01, 384×640 crops, short sides $\{512, 624, 768\}$). Starting from the resulting 78.36-mIoU checkpoint, we warm the event modules for 1,000 iterations and then jointly fine-tune for up to 3,250 iterations at 5×10^{-6} . We use 25/50/100 ms MW, five bins, $\lambda_{\max}=0.03$, $r_{\max}=0.08$, no event dropout, and 500-iteration validation.

5 Discussion and Conclusion

The matched ablations highlight event trust allocation, rather than event availability alone, as a central issue under this protocol. Direct concatenation and an unconstrained adapter reduce daytime accuracy, whereas EventShift improves night mIoU by 4.58 points for a 0.16-point day decrease relative to RGB continuation. This pattern supports anchoring semantics in RGB and applying condition-dependent, bounded corrections. Our implementation places them at **res3/res4**, a middle ground between spatial detail and semantic abstraction; the contribution of injection depth itself is not isolated by the current ablation.

Reliability/support and MW do not provide independent additive gains. Equation (20) and Table 1 show aligned daytime behavior but only partial nighttime overlap. Relative to rel./support-only, full EventShift improves wall and sky IoU by 2.26 and 1.70 points while reducing person and rider IoU by 1.38 and 2.51 points. The mean change across the ten ground-truth-present night classes remains below 0.05 points, showing that a small mean change can mask substantial class redistribution. The separately trained DSEC11 result provides auxiliary evidence for staged adaptation under another taxonomy and protocol. Separately, the official second-place result shows that the broader residual-adaptation principle can contribute to a competitive composite system.

The evidence is limited to one disclosed CoSEC split and a single seed; reliability is an uncalibrated deterministic proxy, and symmetric MW uses 100 ms of look-ahead. Future work will test sequence-disjoint splits, multiple seeds, calibrated reliability, and past-only windows with latency-accuracy analysis. Under the evaluated protocols, bounded, reliability-aware correction delivers substantial nighttime gains while largely preserving daytime accuracy.

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